

Assignment2_20201118_Analysis

Problem 1~4: Using Wine dataset

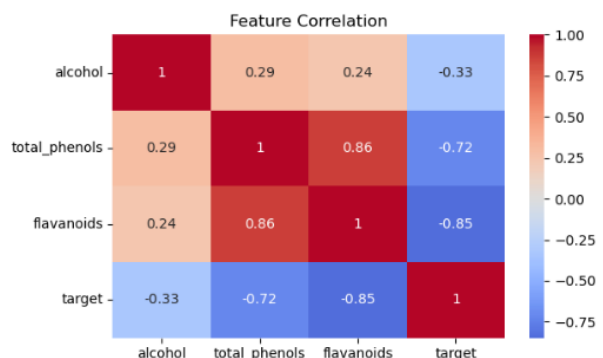
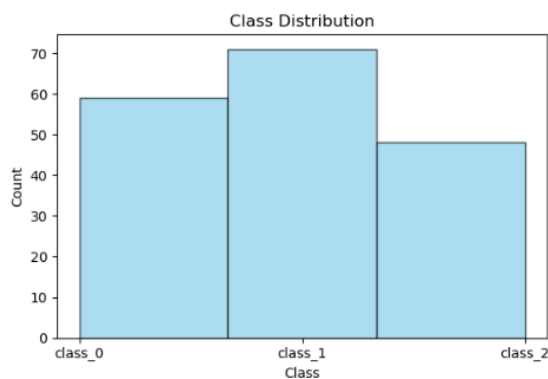
Problem 5: Using Breast Cancer dataset

1. Briefly describe this dataset, focusing on their characteristics.

The dataset used in this assignment is the Wine dataset, which includes 178 samples and 13 continuous features.

It is a multiclass classification problem with three wine classes(class_0, class_1, class_2), containing 59, 71, and 48 samples each, showing a slight imbalance.

The features describe various chemical properties of wine such as alcohol, phenols, and flavonoids. Most values are continuous and vary widely, and some features are highly correlated, for example, total_phenols and flavanoids have a correlation of 0.86.



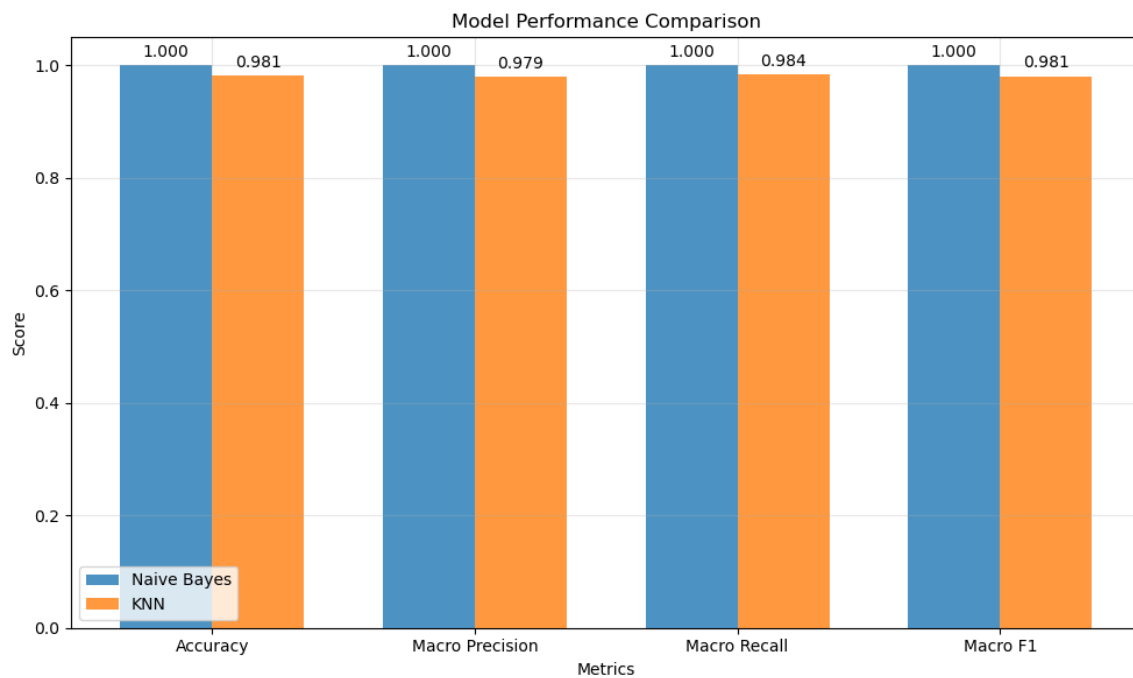
2. Compare the performance of Naive Bayes vs KNN on this dataset. Which performs better and why?

In the Wine dataset, Naive Bayes reached an accuracy of 1.0000, while KNN achieved 0.9815, meaning both models performed very well.

Naive Bayes showed slightly higher accuracy because the dataset is clean and the class boundaries are clear.

KNN captures local structure effectively, but when k increases, the decision boundary becomes smoother, causing minor misclassifications.

Overall, both models performed well, but Naive Bayes was marginally better on this dataset.



3. How do the assumptions of Naive Bayes hold for this dataset?

Naive Bayes assumes that features are conditionally independent and that each class follows a Gaussian distribution.

However, some features in the Wine dataset are strongly correlated, so this independence assumption is not fully satisfied.

Even so, the differences in class means are large, and the feature distributions are relatively simple, so the model still achieved high accuracy despite partial violations of the assumption.

4. How does the choice of distance metric affect KNN performance?

KNN performance can change depending on the distance metric, since it determines how neighbors are selected.

In this experiment, Euclidean, Manhattan, and Minkowski distances were tested, and their performance differences were very small.

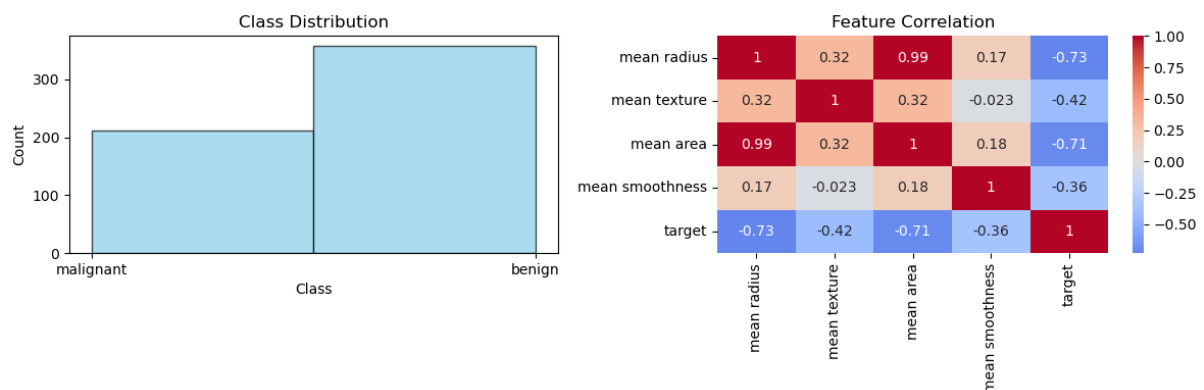
Because the features were standardized and evenly distributed, the choice of distance metric did not significantly affect the results.

Cross-validation results for different metrics all stayed around 0.97 in accuracy, confirming that the normalization step made feature scales uniform and minimized metric sensitivity.

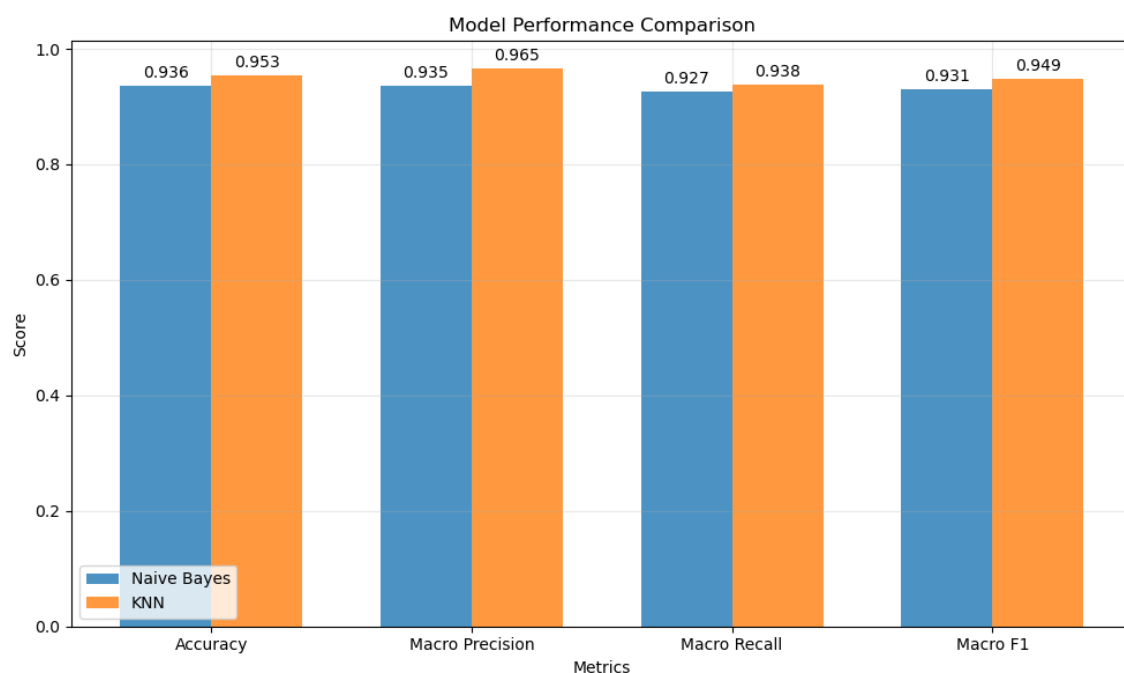
5. Change the dataloader to the breast cancer dataset, by running 1.5 Data Loading

2. Describe how each model's performance changes. From the perspective of dataset, answer why such changes are observed.

After switching to the Breast Cancer dataset, which contains 569 samples and 30 continuous features (212 malignant and 357 benign cases), the model performances changed.



Naive Bayes achieved an accuracy of 0.9357, while KNN reached 0.9532, showing that KNN performed slightly better.



Unlike the Wine dataset where both models classified almost perfectly, the Breast Cancer data is a binary task with more samples and clearer class boundaries.

Since all features were standardized and continuous, the distance-based KNN captured local patterns more effectively.

On the other hand, Naive Bayes assumes feature independence, but the dataset includes

strong correlations among tumor-related features, which reduced its accuracy slightly. Moreover, many features (such as mean radius, mean area, and perimeter) are highly correlated and have skewed distributions, meaning the Gaussian and independence assumptions of Naive Bayes are not fully satisfied.

In contrast, KNN makes no parametric assumption about feature distribution, allowing it to adapt better to these correlations.

As a result, KNN showed more stable and accurate predictions for this dataset.